

TTF Natural Gas Breakout Strategy

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1 Executive Summary

We analysed intraday OHLCV data for TTF Natural Gas (June 2019 – December 2024) to design and evaluate an intraday trend-following breakout strategy.

Our analysis identifies that breakouts of the first-hour high and low, conditioned on elevated first-hour volatility, display strong positive payoff asymmetry: most trades are small losses, but a small fraction generate large profits. This implies that the strategy’s edge comes from rare, outsized winners rather than frequent small gains. Consequently, we do not use fixed take-profit levels and instead hold positions until end-of-day.

At the same time, adverse excursions display heavy left tails. This motivates explicit downside control. We introduce (i) a hard stop at 1.25σ below the breakout level, chosen to sit above typical noise but below extreme adverse moves, and (ii) an early structure-failure rule that exits trades which revert back inside the opening range by 0.1σ within 30 minutes. Path-based, exit-invariant diagnostics confirm that this early reversion identifies a low-quality regime with low MFE and high MAE, and that exiting such trades does not materially destroy latent winners.

The resulting strategy achieves a cumulative performance of 45.7σ , an annualized Sharpe ratio of 1.15σ , a maximum drawdown of -17.7σ , and a trade win rate of 26.7%.

2 Background

Natural Gas Crisis. We define a crisis regime spanning January 2021 to July 2023, corresponding to the period of exceptional volatility following the European energy shock. We report results both within and outside this regime to assess robustness, and treat the non-crisis period as the more representative baseline for long-run behavior.

Gap-Aware Volatility Measurement & Data Availability Normalization. An initial data audit reveals substantial missingness, concentrated in the first trading hour and in the pre-crisis period (Figure 3). To avoid introducing artifacts through interpolation or forward-filling, we restrict attention to observed returns only and construct a gap-aware volatility estimator.

Specifically, we compute volatility from log-returns normalized by the time elapsed between observations, and exclude returns that span more than 10 minutes or cross trading sessions:

$$\text{Volatility} = \frac{\log(\text{return})^2}{\Delta \text{ minutes}}.$$

In addition, we enforce a minimum data coverage threshold of 20% per day to ensure comparability across days and to prevent high-activity or densely sampled days from disproportionately influencing the analysis.

Volatility Normalization. All reported MFE, MAE, PnL, Sharpe ratios, and drawdowns are normalized by first-hour volatility in order to make outcomes comparable across days with different absolute price levels and volatility regimes. Except where explicitly stated (notably in the exploratory analysis), references to σ or volatility refer exclusively to this first-hour estimate.

Code Repository. All code is available at: GitHub repository

3 Exploratory Data Analysis

Intraday Volatility. Intraday volatility is strongly front-loaded: the opening minutes account for a disproportionate share of daily price variance, particularly on Mondays and during the crisis regime. Volatility then compresses rapidly over the first hour, followed by a modest re-expansion mid-day and a smaller rise approximately 30 minutes before the close, before compressing again into the close. Across all time buckets, the mean exceeds the median, indicating positively skewed volatility with rare but extreme high-volatility days (Figures 4–5).

Intraday Volume. Volume displays a similar front-loaded structure, with elevated activity at the open, most pronounced on Mondays and outside the crisis regime, followed by a low-activity period after the first hour. A large and persistent spike occurs approximately 30 minutes before the close, especially on Fridays, before volume collapses into the close. As with volatility, volume distributions are right-skewed, with the mean exceeding the median throughout the day, reflecting the presence of occasional extreme activity days (Figures 6–7).

Expansion Ratios. Opening-range expansion ratios show a median expansion potential of approximately 2.25 for 60-minute ranges and 2.75 for 30-minute ranges, supporting the economic plausibility of breakout strategies based on early price ranges. Expansion ratios are slightly lower on Mondays and during the crisis period, suggesting more frequent false starts or choppy early trading. The mean consistently exceeds the median, indicating that while most days display limited expansion, a small fraction of days generate very large moves (Figures 10–13).

Conclusion. Volatility and volume are front-loaded and positively skewed, and expansion ratios display heavy right tails. Most days do not expand materially beyond the opening range, but a small subset produce large directional moves. This implies that a viable strategy must tolerate frequent small losses, avoid capping profits, and remain exposed to rare but large outcomes. Lower median but fatter-tailed expansion during crisis periods and on Mondays further suggests a higher incidence of false breakouts alongside more extreme valid ones, motivating volatility conditioning and fast failure detection mechanisms.

4 Trend Persistence Analysis

First Order Analysis. We evaluate candidate breakout levels by their volatility-normalized path asymmetry (Median MFE - Median MAE). Positive asymmetry indicates that, conditional on a level being crossed, the subsequent price path display directional persistence rather than mean reversion. Thus the objective is to identify levels that produce statistically meaningful separation of price paths into favorable and adverse regimes. We retain only levels satisfying the following criteria: (i) a minimum sample size of $n \geq 600$ to ensure statistical reliability and reduce the risk of overfitting under higher-order conditioning. (ii) regime robustness, requiring that asymmetry does not change sign across crisis and non-crisis samples. and (iii) a minimum median asymmetry of 0.15 in both regimes to exclude levels whose edge is negligible.

Table 1 reports the results. Levels highlighted in green satisfy all selection criteria and are retained for further analysis and cells highlighted in red indicate violations of the corresponding filters and motivate exclusion.

Confluence Analysis. Combining multiple levels fails to improve payoff asymmetry or mitigate downside risk and frequently reduces both sample size and signal strength. As a result, confluence-based specifications are discarded from subsequent analysis (Table 2).

Second Order Analysis. Conditioning on first-hour volatility materially improves payoff asymmetry and reduces tail risk across regimes. We therefore restrict further analysis to breakouts of the opening range that occur on high-volatility first-hour days. (Table 3)

Level	Dir	n	MFE	MAE	Q90 MAE	Asym - Crisis	Asym - Non-Crisis
ATR + 1R	down	81	1.282	0.579	2.484	0.725	0.595
ATR + 0.5R	up	344	1.141	0.848	2.703	0.266	0.401
ATR + 0.25R	down	585	1.071	0.866	3.053	0.059	0.290
VWAP + 2σ	up	377	1.158	0.975	3.327	0.107	0.258
OR LOW	down	640	1.147	0.887	3.204	0.188	0.248
ATR - 0.75R	down	139	0.973	0.821	1.985	0.059	0.248
VWAP	up	806	1.381	0.980	3.071	0.440	0.247
VWAP - 0.5σ	down	740	1.233	0.981	3.368	0.157	0.224
VWAP + 1σ	down	604	1.158	0.975	3.221	0.054	0.207
ATR + 0.75R	down	164	0.930	0.764	2.605	0.283	0.186
ATR - 0.25R	down	541	1.071	0.866	3.100	0.268	0.177
VWAP + 0.5σ	down	742	1.233	0.981	3.324	0.110	0.156
VWAP + 0.5σ	up	785	1.335	1.096	3.218	0.229	0.155
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OR HIGH	up	697	1.270	1.084	3.204	0.274	0.057

Table 1: Breakout Level Analysis

Level	Dir	n	MFE	MAE	Q90 MAE	Asym - Crisis	Asym - Non-Crisis
OR LOW $\wedge$ VWAP - 0.5σ	down	624	1.133	0.887	3.206	0.220	0.235
OR HIGH $\wedge$ VWAP + 0.5σ	up	670	1.241	1.107	3.273	0.271	0.020
OR HIGH $\wedge$ VWAP	up	604	0.991	1.182	3.318	0.026	-0.223

Table 2: Confluence Analysis

Third Order Analysis. While third-order conditioning highlights intuitive structural effects (e.g., better tail behavior for late or expansion-regime breakouts), it fails to improve asymmetry relative to the second-order high-volatility baseline. Moreover, it often reduces median asymmetry and significantly lowers statistical power through sample size reduction, without offsetting risk benefits. We therefore exclude third-order conditioning from the final strategy. (Table 7)

5 Strategy Design

Take-Profit Rule. The payoff distribution is strongly right-skewed: most trades incur small losses, while a small fraction generate large gains. In addition, the time-to-MFE distribution is widely dispersed, indicating that profitable trends can take a long time to develop. These two observations imply that the strategy’s edge depends on remaining exposed to rare, extended moves rather than harvesting frequent small profits. Consequently, we do not impose fixed take-profit levels and instead exit positions only at end-of-day.

Hard Stop-Loss Rule. Second-order analysis reveals heavy left tails in adverse excursions, with a median MAE of approximately $0.8\text{--}0.9\sigma$ and a 90th percentile around 2.9σ . This motivates a hard stop placed at 1.25σ beyond the breakout level: far enough to avoid frequent stop-outs from normal noise, yet close enough to limit exposure to extreme adverse moves.

Early Structure-Failure Stop. Economic intuition suggests that genuine breakouts should not immediately revert back into the opening range. We therefore introduce an early exit rule: if price re-enters the range by 0.1σ within the first 30 minutes after entry, the trade is exited. This rule is intended to identify and remove false breakouts early, before they accumulate large adverse excursions.

Level	Dir	Vol	n	MFE	MAE	Q90 MAE	Asym(C)	Asym(NC)
OR LOW	down	Low	323	1.211	1.120	3.603	-0.121	0.213
OR LOW	down	High	317	1.053	0.783	2.895	0.275	0.555
OR HIGH	up	Low	367	1.233	1.160	3.658	0.269	-0.014
OR HIGH	up	High	330	1.287	0.918	2.897	0.275	0.546
VWAP $+0.5\sigma$	up	Low	404	1.383	1.205	3.578	0.327	0.130
VWAP $+0.5\sigma$	up	High	381	1.229	1.045	2.975	0.188	0.207
VWAP -0.5σ	down	Low	380	1.304	1.145	3.724	0.007	0.224
VWAP -0.5σ	down	High	360	1.185	0.870	2.880	0.308	0.217
VWAP	up	Low	406	1.499	1.143	3.445	0.604	0.231
VWAP	up	High	400	1.248	0.823	2.792	0.375	0.438

Table 3: Second Order Analysis, Conditioned on First Hour Volatility

Validation of Stop Rules. The barrier-race analysis (Figure 1) shows that breakout persistence is scale-dependent: small favorable barriers are typically reached quickly when a breakout is genuine, while false breakouts tend to fail early. This supports the use of a fast structure-failure rule and confirms that early follow-through is a meaningful signal of breakout quality.

We further assess whether the stop rules destroy latent winners. The latent-winner analysis shows that while some trades stopped early would eventually reach favorable levels, the majority of them do not become large winners.

Path comparisons (Tables 4 and 5) confirm that trades stopped by either the early or hard stop display strongly negative MFE/MAE asymmetry, validating that these rules primarily remove low-quality paths.

Finally, a sweep over alternative hard-stop distances shows that tighter stops can increase cumulative PnL and Sharpe, but at the cost of significantly worse drawdowns. We therefore retain the 1.25σ stop as a conservative choice that prioritizes tail-risk control over marginal improvements in return metrics. (Tables 5-6)

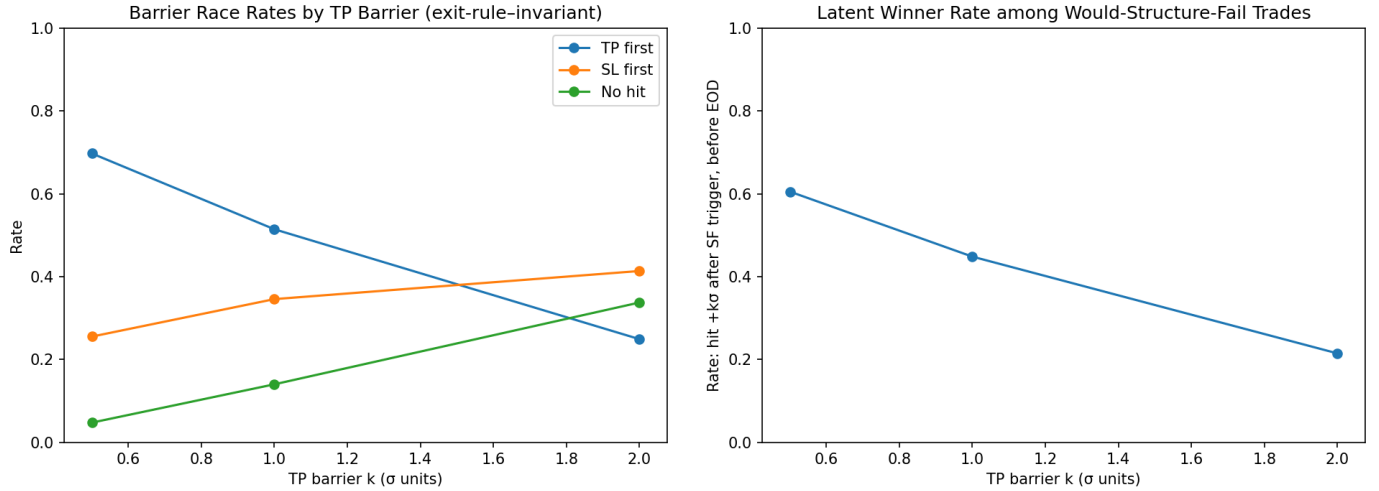


Figure 1: Barrier Race & Latent Winner Analysis

Group	n	MFE	MAE	Asym
Would-structure-fail	261	0.823	1.347	-0.524
No-structure-fail	225	1.434	0.557	0.877

Table 4: Path Comparison Early Structure Failure Stop to EOD

σ_{stop}	Group	n	MFE	MAE	Asym
0.75	Would-hit-hard-stop	296	0.768	1.700	-0.932
0.75	No-hard-stop-hit	187	1.931	0.295	1.636
1.00	Would-hit-hard-stop	250	0.661	1.933	-1.272
1.00	No-hard-stop-hit	233	1.761	0.408	1.354
1.25	Would-hit-hard-stop	207	0.560	2.160	-1.600
1.25	No-hard-stop-hit	276	1.717	0.460	1.257

Table 5: Path Comparison Hard Stop to EOD.

σ_{stop}	PnL	Sharpe	MaxDD
0.75	46.701	1.246	-22.448
1.00	41.292	1.070	-22.080
1.25	42.099	1.045	-19.369

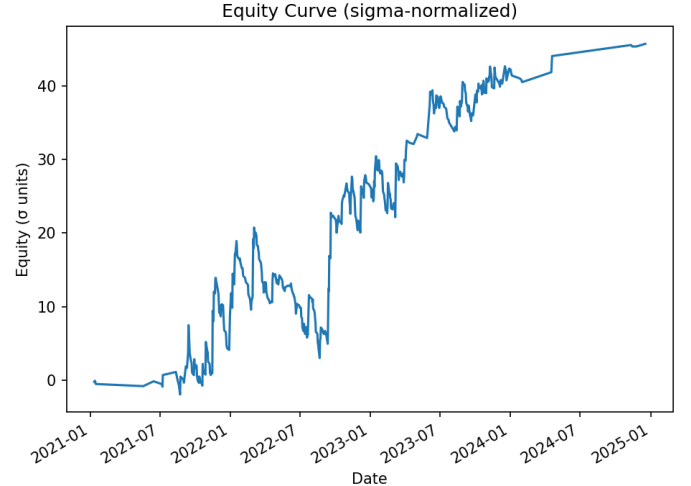
Table 6: Strategy-Level Performance by Hard Stop Distance.

6 Evaluation & Limitations

The strategy displays meaningful performance with positive skew, but high drawdowns, and robustness across market regimes, suggesting that the edge is not driven by a narrow set of extreme observations. However, the return distribution remains highly concentrated in a small number of large winners, implying that performance is sensitive to sample length and may be unstable over shorter horizons. In addition, transaction costs, slippage, and liquidity constraints have not been modeled and may materially reduce realized performance, particularly given the intraday execution and stop-driven exits. These factors suggest that while the results are meaningful, the strategy should be treated as exploratory and would require out-of-sample validation, stress testing, and live monitoring before any capital deployment.

Metric	Value
Sample (sigma days)	1025
Trading days / trades	486 / 486
Total PnL (cumulative, σ)	45.686
Sharpe (daily σ , annualized)	1.149
Max drawdown (σ)	-17.722
Win rate (trades)	0.267
Exit reason: structure fail (30m)	260
Exit reason: hard stop (-1.25σ)	73
Exit reason: end-of-day	153
PnL by side (σ): long / short	21.923 / 23.762
PnL by regime (σ): crisis / non-crisis	35.041 / 10.644
Trade PnL (σ): med / p90 / max	-0.208 / 1.709 / 8.498
Hold dur(min): med / p75 / max	23.5 / 421.5 / 537
Entry (min): med / p75 / max	34.5 / 96.75 / 535

(a) Backtest evaluation (σ units)



(b) Equity curve (PnL in σ units)

Figure 2: Strategy backtest performance summary

Outlook & Future Work A notable diagnostic result is that trades hitting the hard stop, especially at the 1.25σ level, display a much stronger and more stable asymmetry than the breakout trades themselves. This suggests that stop events may identify a distinct price regime rather than merely controlling risk, motivating future research into failure-conditioned or reversal strategies.

Transaction costs, liquidity constraints, slippage as well as high-friction execution should also be modeled.

Finally, the strategy currently uses implicit unit sizing. Future work should incorporate explicit position sizing and risk budgeting to control drawdowns and tail exposure.

A Additional Figures & Tables

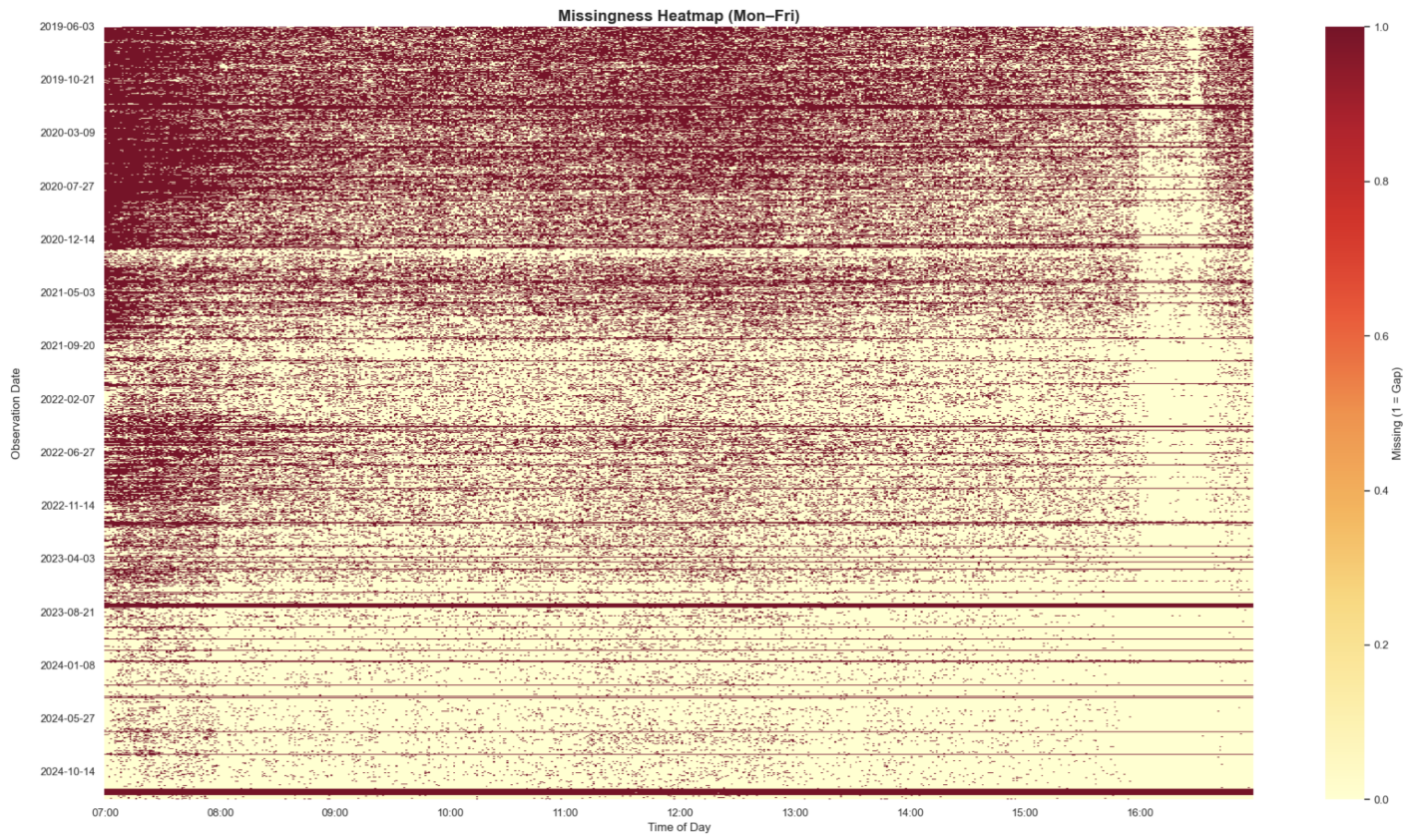


Figure 3: Data Missingness Heatmap

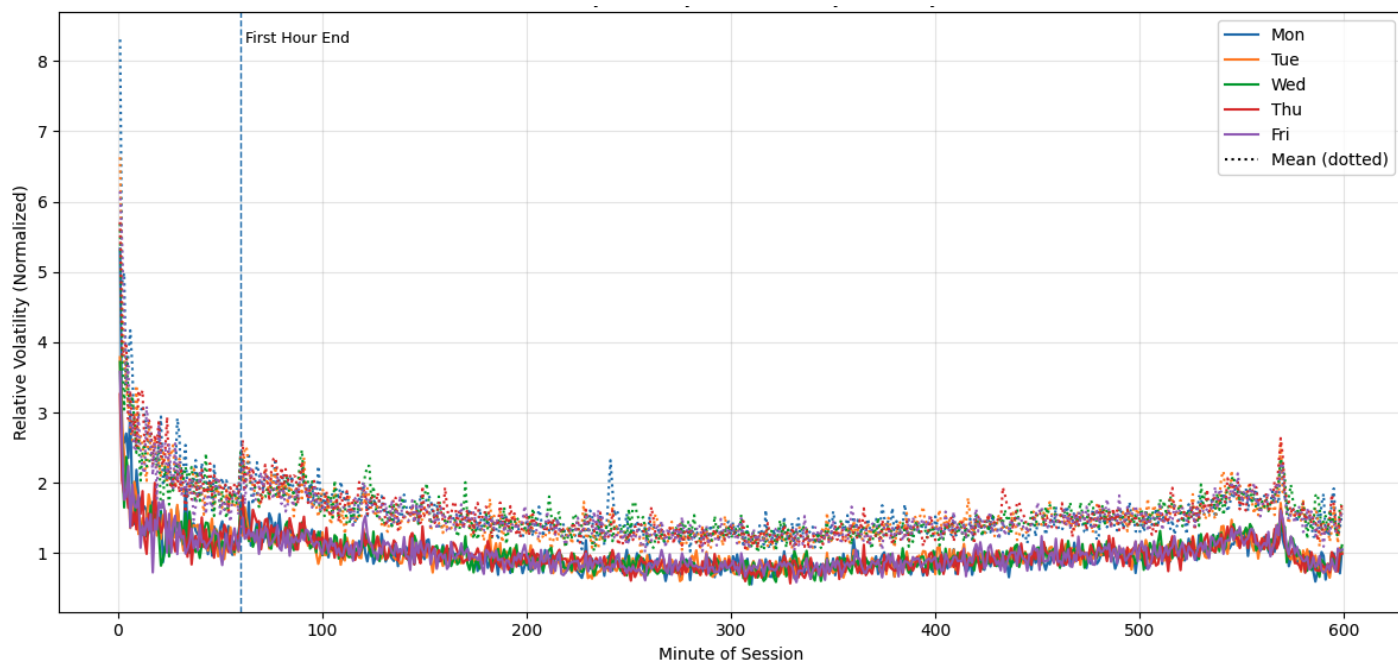


Figure 4: Intraday Volatility Curve by Weekday (1m) (Median)

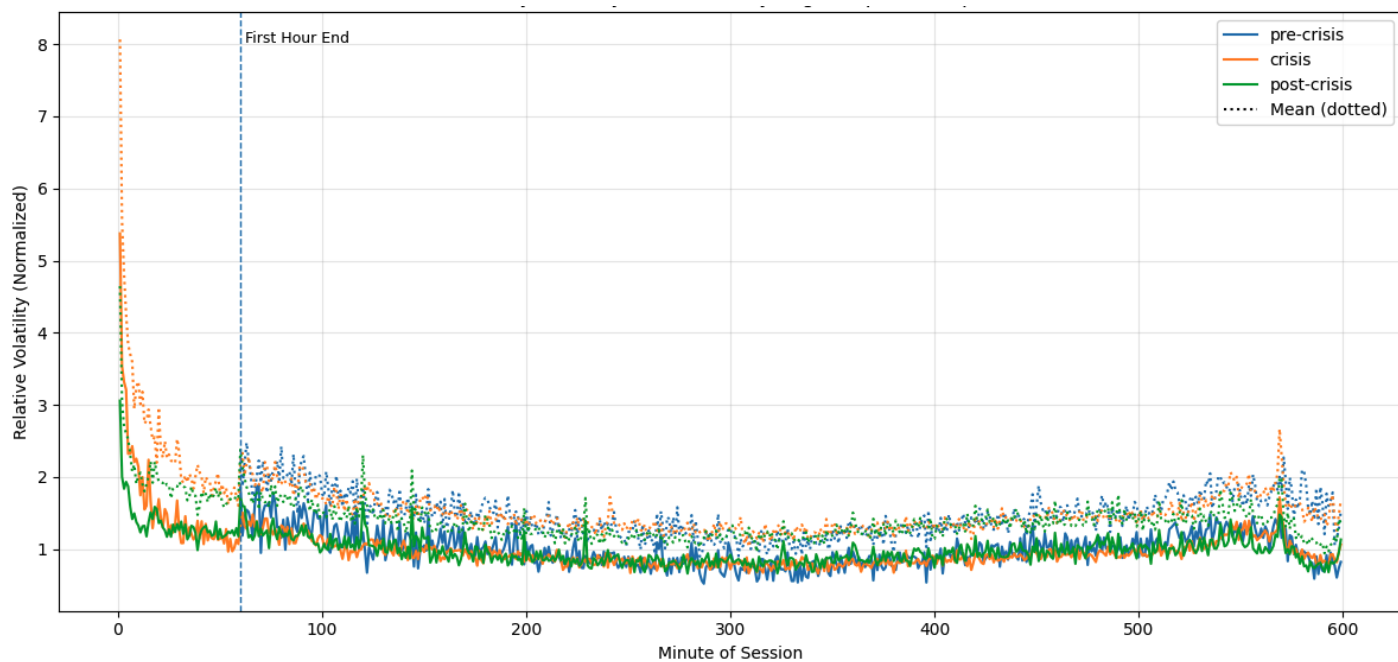


Figure 5: Intraday Volatility Curve by Regime (1m) (Median)

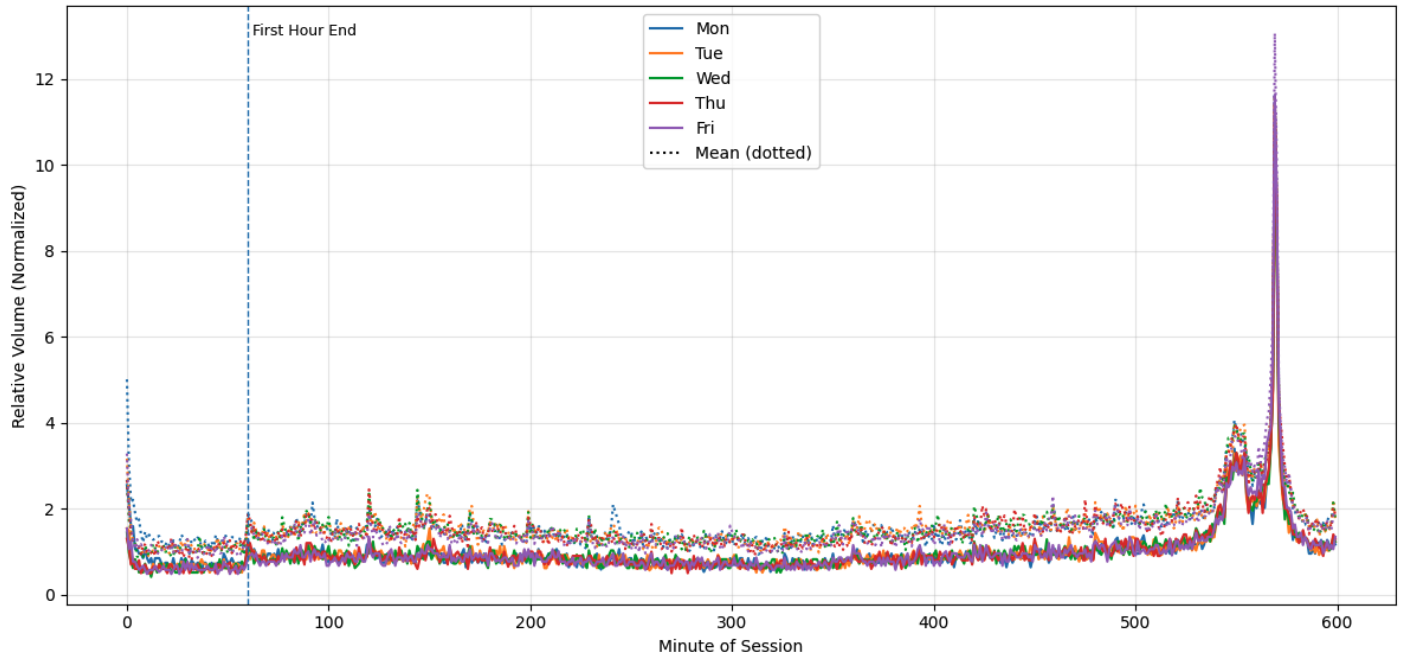


Figure 6: Intraday Volume Curve by Weekday (1m) (Median)

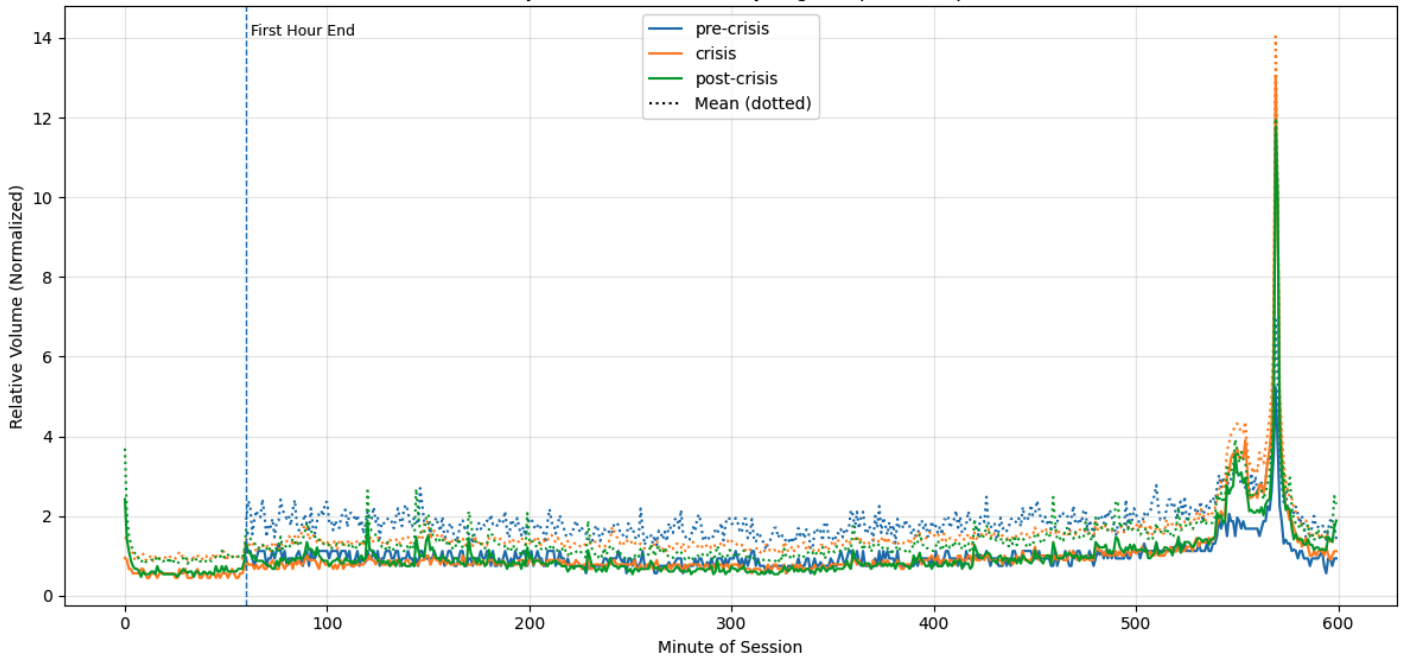


Figure 7: Intraday Volume Curve by Regime (1m) (Median)

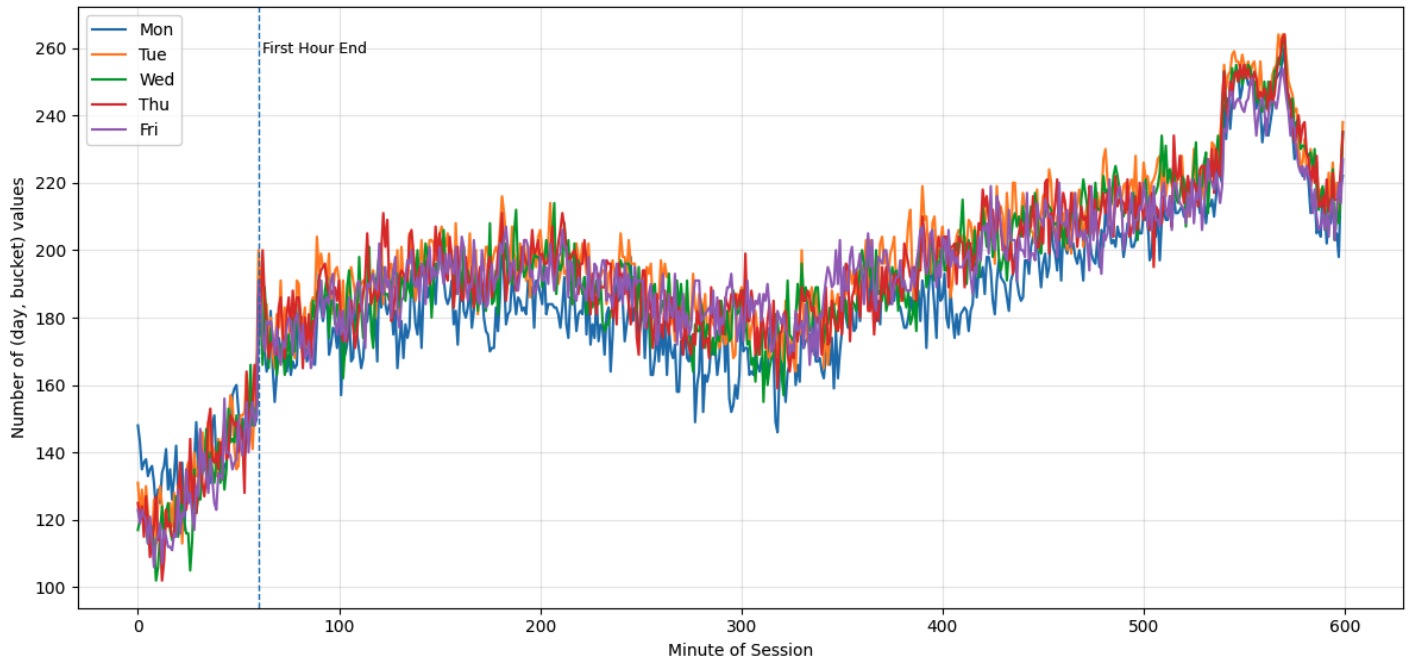


Figure 8: Coverage by Weekday: Contributing Day-Count per Minute

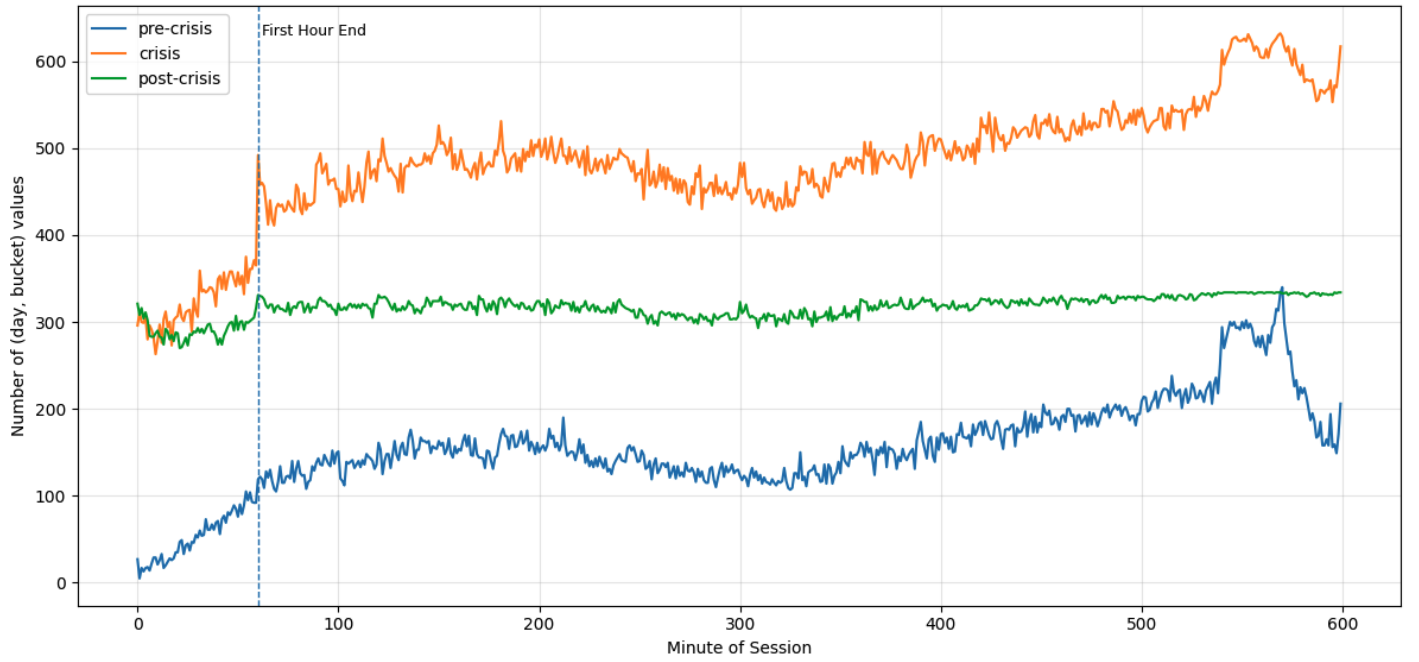


Figure 9: Coverage by Regime: Contributing Day-Count per Minute

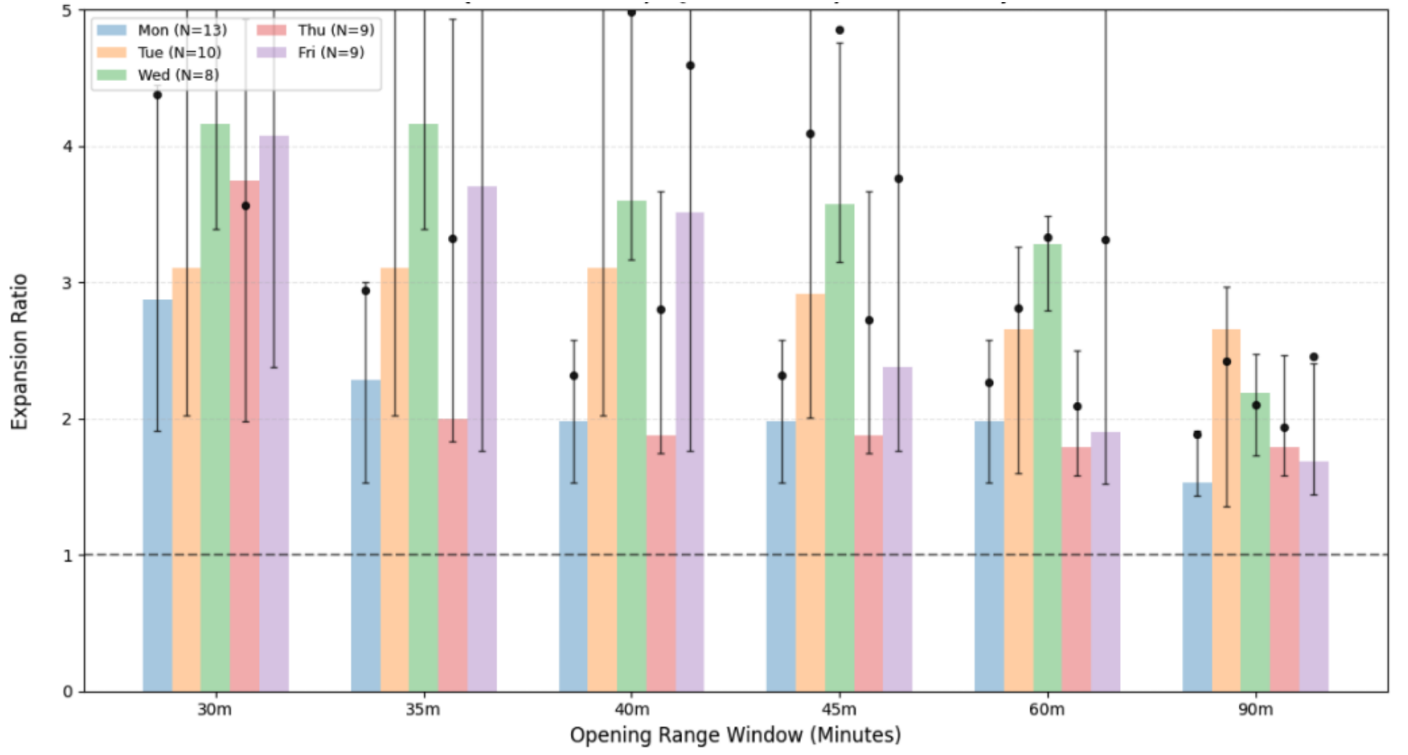


Figure 10: Expansion Ratio by Weekday (Pre-Crisis) (Bars=Median, IQR=P25-P75, Dots=Mean) (Unstable)

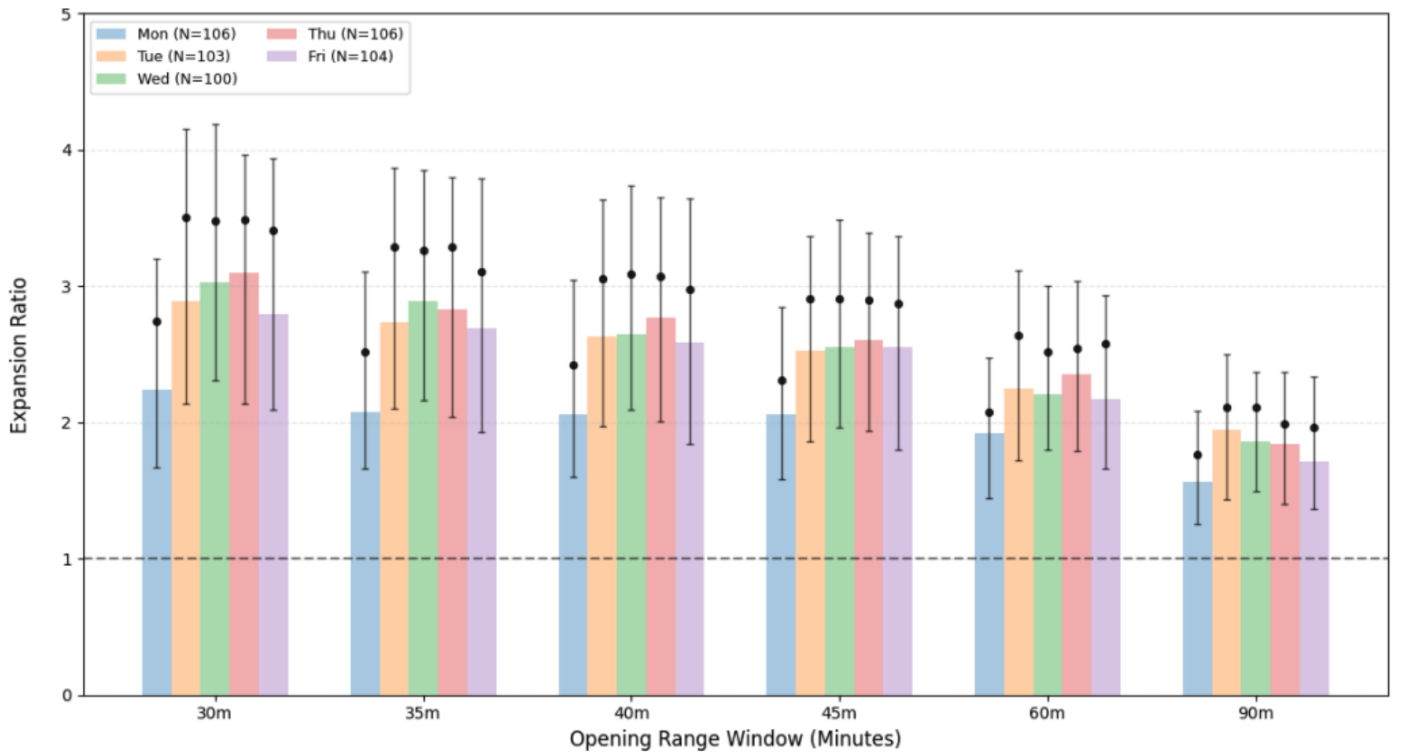


Figure 11: Expansion Ratio by Weekday (Crisis) (Bars=Median, IQR=P25-P75, Dots=Mean)

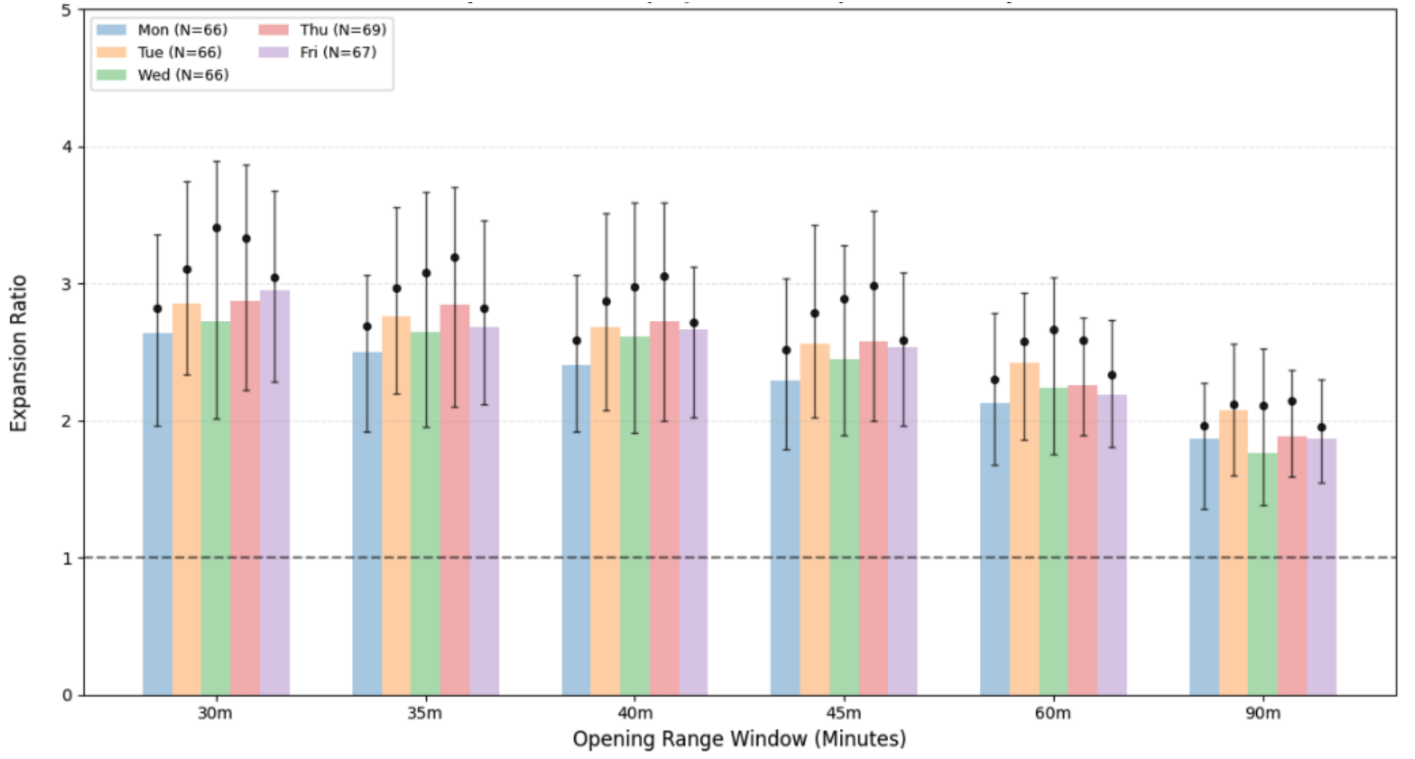


Figure 12: Expansion Ratio by Weekday (Post-Crisis) (Bars=Median, IQR=P25-P75, Dots=Mean)

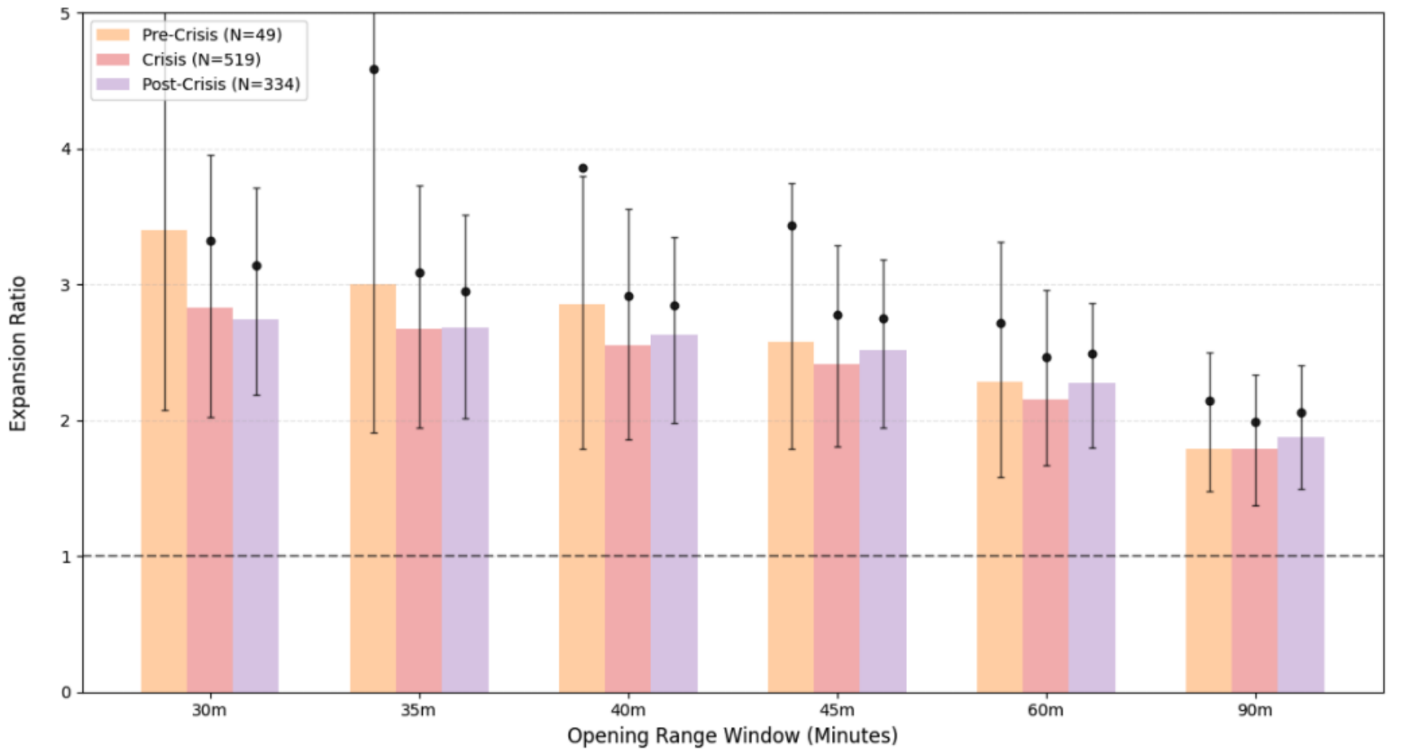


Figure 13: Expansion Ratio by Regime (Bars=Median, IQR=P25-P75, Dots=Mean) (Unstable Pre-Crisis)

Level	Dir	Vol	Feature	Bucket	n	MFE	MAE	Q90(MAE)	Asym
OR HIGH	up	High	Range	Compression	168	1.299	0.801	2.440	0.479
OR HIGH	up	High	Range	Expansion	162	1.268	1.108	3.116	0.253
OR HIGH	up	High	Dist at Cross	Large	162	1.423	1.015	2.878	0.295
OR HIGH	up	High	Dist at Cross	Small	168	1.254	0.889	2.725	0.343
OR HIGH	up	High	Time at Cross	Early	172	1.602	1.150	3.267	0.393
OR HIGH	up	High	Time at Cross	Late	158	1.066	0.635	2.248	0.292
OR HIGH	up	High	Drift Align	Aligned	196	1.332	1.009	3.248	0.265
OR HIGH	up	High	Drift Align	Not aligned	134	1.218	0.819	2.250	0.492
OR HIGH	up	High	Volume Confirm	High	173	1.163	0.872	2.877	0.311
OR HIGH	up	High	Volume Confirm	Low	157	1.422	0.977	2.740	0.336
OR LOW	down	High	Range	Compression	156	0.970	0.787	3.176	0.144
OR LOW	down	High	Range	Expansion	161	1.148	0.770	2.535	0.389
OR LOW	down	High	Dist at Cross	Large	161	1.178	0.910	2.860	0.399
OR LOW	down	High	Dist at Cross	Small	156	0.894	0.669	3.010	0.168
OR LOW	down	High	Time at Cross	Early	154	1.089	1.062	3.531	0.168
OR LOW	down	High	Time at Cross	Late	163	0.971	0.565	1.985	0.398
OR LOW	down	High	Drift Align	Aligned	180	1.050	0.766	2.954	0.289
OR LOW	down	High	Drift Align	Not aligned	137	1.081	0.789	2.873	0.361
OR LOW	down	High	Volume Confirm	High	152	1.201	1.123	3.220	0.125
OR LOW	down	High	Volume Confirm	Low	171	1.212	1.116	3.984	0.084

Table 7: Third-order conditioning within the selected high-volatility regime